Harsh N. Patil

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Education

Indian Institute of Technology Madras Chennai, India
B.S. in Data Science and Applications | CGPA: 6.92 | Project CGPA: ***9.5*** *May 2026*

Technical Skills

Languages: Python, Java, C, JavaScript
ML / Data Science: TensorFlow, Keras, PyTorch, OpenCV, Scikit-learn, NumPy, SciPy, Pandas, Matplotlib
Frameworks & Tools: Flask, Streamlit, VueJS, Docker, PostgreSQL, Supabase, CI/CD, Git, Jira
Domains: Computer Vision, Machine Learning, SaaS Architecture, DevOps, Deep Learning, Transfer Learning
Certifications: Anthropic MCP Architecture, NPTEL Consumer Psychology, NPTEL Soft Skill Development

Projects

Multi-Tenant Solar CRM & Document Automation SaaS | *Flask, Docker, Supabase, CI/CD* Jun 2026

- Designed and developed a multi tier SaaS platform using Flask for solar agencies to manage clients, utilizing Supabase storage buckets to securely handle sensitive documents and eliminate database bloat.
- Solved critical server memory issues by rewriting the asynchronous PDF generation engine, replacing Playwright with WeasyPrint to cut RAM usage by 70% and ensure reliable server uptime on budget constraints.
- Created custom Docker containers and a GitHub Actions pipeline to fully automate testing, secure SSH deployments, and execute rolling server restarts without any downtime.
- Configured a DigitalOcean production server with virtual swap space to prevent crashes under heavy traffic, and integrated Cloudflare to automatically encrypt sensitive financial files.

Pneumonia Detection Using Chest X-Rays | *Python, TensorFlow, Keras, OpenCV* Jan 2026

- Created a clinical decision support system using Convolutional Neural Networks to accurately classify chest X-ray images into normal or pneumonia categories to assist healthcare professionals.
- Built a reliable image preprocessing pipeline using OpenCV and NumPy for resizing and contrast enhancement, and applied data augmentation techniques to prevent overfitting on limited medical datasets.
- Utilized transfer learning with prebuilt architectures like ResNet50 and VGG16 to improve feature extraction, specifically optimizing for high recall to minimize missed diagnoses.

Crop Yield Prediction Machine Learning Pipeline | *Python, Scikit-learn, Pandas, XGBoost* May 2026

- Developed a complete machine learning pipeline to predict crop yields by analyzing soil, climate, and fertilizer features across a large agricultural dataset.
- Evaluated multiple predictive models including XGBoost and Random Forest through rigorous cross validation, achieving high accuracy and explaining over 91% of the variance in yield.
- Conducting statistical testing and feature importance analysis to identify the most critical environmental variables driving crop production.
- Structured the project into reproducible scripts for data preprocessing, model training, and evaluation to allow for easy future deployment and scalability.

Experience

Full Stack Developer, Pariman (Practice Management Platform) Jalgaon, India
Chartered Accountancy Firm *Jul 2026 – Present*

- Built an internal ERP replacing spreadsheet based workflows for a CA practice, covering client onboarding, statutory compliance filings (ITR, GST, TDS, ROC, Tax Audit, RERA), staff attendance, payroll, leave and expense management.
- Architected a permission based access control system with per (module, tab, action) grants across 25 modules, replacing role string checks, including a document access vault with approval workflows, expiring grants, and audit logging.
- Designed a universal compliance vault aggregating every filed deliverable per client across services, with per service upload limits, DLP controls (view only for unprivileged staff), and encrypted credential storage using Fernet.
- Built on Python/Flask, PostgreSQL, SQLAlchemy with Alembic, React/Vite, Docker Compose, nginx, and gunicorn, with JWT auth using HTTP only cookies and CSRF protection; integrated WhatsApp Cloud API and SMTP for client document dispatch and OTP verification, plus Google Calendar sync.
- Scaled to roughly 59k lines of code across 25 API blueprints, 72 data models, 83 migrations, and 99 React components, backed by a 283 case pytest suite.

Achievements & Leadership

National Finalist, HCL Guvi Hackathon at the AI Summit 2026, held at Bharat Mandapam, New Delhi (Jan 2026).

Finalist, Syngenta Hackathon at IIT Madras BS Paradox technical fest (2026).

Public Relations Head, AAYAM Club, IIT Madras BS (2024).

Deputy Head of Dramatics Society, IIT Madras (2024).